When Similar Means Different: Evaluating LLMs on Arabic–Hebrew Cognates

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Abstract

Arabic and Hebrew, as closely related Semitic languages, share many words with similar surface forms, including true cognates, false friends, and modern loanwords. These lexical relationships create ambiguity for cross-lingual semantic interpretation, as similar forms may correspond to shared, divergent, or borrowed meanings. To evaluate whether LLMs can make these distinctions, we introduce SemCog Bench, a curated benchmark of 1,858 Arabic–Hebrew word pairs with sentence-level annotations for cognate identification and semantic disambiguation. We evaluate a diverse set of open-source and proprietary LLMs across multiple input representations and contextual settings. Our results show reliance on surface-form similarity, with weaker performance on false friends and wide variation on loanwords. We further find that context and scale yield model-dependent gains, while original-script inputs generally perform best. Our findings reveal limitations in cross-lingual form-meaning reasoning and establish SemCog Bench as a benchmark for cross-lingual lexical semantics. Our code and data are publicly available.[1]

1 Introduction

Arabic and Hebrew, as closely related Semitic languages, exhibit substantial lexical overlap, including many words with similar surface forms. While some of these words are *true cognates* that preserve semantically related meanings across the two languages, others are *false friends* that exhibit similar orthographic and phonological forms but differ in meaning (Cahyawijaya et al., 2025). A third category consists of *loanwords* borrowed from other languages, which may also appear highly similar in Arabic and Hebrew. This lexical overlap creates substantial ambiguity, since similar forms across Arabic and Hebrew may differ in both lexical relationship and meaning.

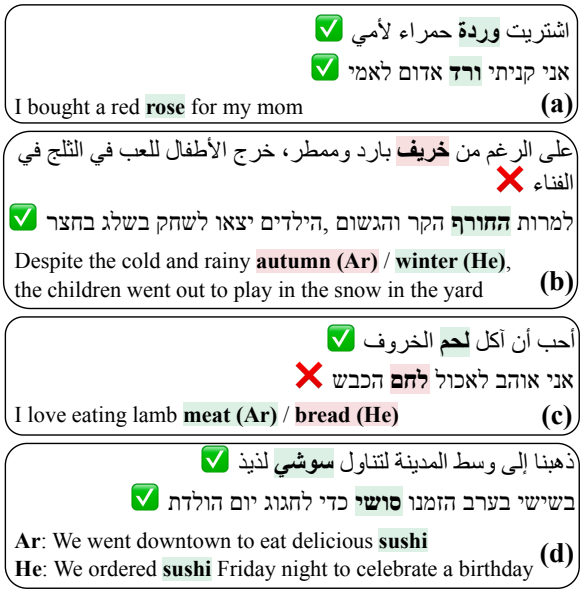


Figure 1: Examples of Arabic and Hebrew true cognates (a), false friends (b, c) and loanwords (d). Correct usages are marked in green and incorrect usages in red.

Figure 1 illustrates examples of these lexical relationships. True cognates, such as the Arabic word وردة (*wrdħ*[2], 'rose') and the Hebrew word ורד (*wrd*[3], 'rose'), share similar forms and meanings across the two languages (Figure 1(a)). In contrast, false friends preserve strong surface similarity despite semantic divergence; for example, the Arabic word خريف (*xryf*, 'autumn') closely resembles the Hebrew word חורף (*xwrp*, 'winter') (Figure 1(b)). Loanwords may likewise appear similar, but for a different reason: سوشي (*swšy*, 'sushi') and סושי (*swšy*, 'sushi') (Figure 1(d)) are both modern borrowings rather than inherited Semitic cognates. These cases demonstrate the different lexical relationships that can underlie similar forms across Arabic and Hebrew.

[1] https://github.com/mbzuai-nlp/SemCog

[2] Arabic HSB transliteration (Habash et al., 2007).

[3] Hebrew transliteration following Shilon et al. (2010).

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In this work, we introduce SemCog Bench, a curated benchmark of 1,858 Arabic–Hebrew true cognates, false friends, and loanwords with sentence-level annotations for cognate identification and semantic disambiguation. Using SemCog Bench, we evaluate a diverse set of LLMs, including open-source and proprietary multilingual models as well as Arabic- and Hebrew-centric models. We address four research questions: **RQ1:** How accurately do LLMs perform Arabic–Hebrew cognate identification and semantic disambiguation? **RQ2:** How does input representation (raw, diacritized, Romanized, or phonetic) affect performance? **RQ3:** Does sentence-level context improve cognate identification? **RQ4:** To what extent does model scaling improve performance on SemCog Bench? Our main contributions are as follows:

1. We introduce SemCog Bench, a benchmark of Arabic–Hebrew true cognates, false friends, and loanwords for cognate identification and semantic disambiguation.
2. We evaluate a broad range of LLMs using SemCog Bench, across input representations, context settings, and model scales.
3. We show that LLMs exhibit reliance on surface-form similarity, struggling to distinguish false friends from true cognates, while loanword performance varies across models.
4. We find that context and scale provide model-dependent gains, and that original-script input generally outperforms other representations.

2 Background and Related Work

2.1 Arabic–Hebrew Linguistic Background

Arabic and Hebrew are closely related Semitic languages characterized by rich morphology and orthographic ambiguity. Both employ root-and-pattern word formation, where words are derived by combining a consonantal root, often triconsonantal, with vocalic patterns, resulting in large and sparse vocabularies. In addition, both languages use optional diacritics to represent short vowels–*tashkil* in Arabic and *nikkud* in Hebrew–that are typically omitted in modern text (Habash, 2010; Seddah et al., 2013). As a result, many surface forms are ambiguous, posing challenges for computational modeling (Tsarfaty et al., 2020).

At the same time, the shared Semitic ancestry of Arabic and Hebrew gives rise to numerous lexical parallels, including words with similar orthographic or phonological forms. These include true cognates, false friends, and loanwords. Cognates are words in different languages that share a common historical origin (Lijewska, 2020; Abu-Rabiah, 2025). When cognates preserve both form and meaning across the two languages, they constitute true cognates (Figure 1(a)); when their forms are similar but their meanings diverge, they may result in false friends (Chamizo Domínguez and Nerlich, 2002). Hebrew לחם (*lxm*, 'bread') and Arabic لحم (*lHm*, 'meat'), for example, both derive from the Semitic root *l-H-m* yet denote different things (Figure 1(c)). Loanwords (Figure 1(d)), instead, owe their similarity to parallel borrowing rather than common descent. Although the term generally refers to words borrowed from one language into another (Mahajna, 2019), we use it specifically for forms borrowed from languages that do not share the relevant inherited Semitic root. Distinguishing these lexical relationships requires integrating orthographic, phonological, and semantic evidence rather than relying on surface similarity alone.

2.2 Modeling Cognate Relations

Computational modeling of cognate relations has traditionally focused on identifying historically related words across languages. Early work relied on orthographic and phonological similarity-based methods (List, 2012b,a; Hauer and Kondrak, 2015), later incorporating probabilistic, clustering, and feature-based techniques (Rama, 2018; Rama and List, 2019; MacSween and Caines, 2020; Cristea et al., 2021). More recent work has incorporated neural architectures and cross-lingual representations (Kanojia et al., 2020b,a, 2021; Bafna et al., 2022; Nath et al., 2022; Akavarapu and Bhattacharya, 2023, 2024; Uban et al., 2025).

Alongside these methods, a range of resources has been developed for cognate identification (Batsuren et al., 2019; Cristea et al., 2021; Batsuren et al., 2022; List et al., 2022). Two efforts are particularly close to our work. Dinu et al. (2023) introduced a benchmark spanning both cognates and borrowings across Romance languages; like ours, their taxonomy treats borrowing as distinct from inheritance rather than collapsing the two. Cahyawijaya et al. (2025) introduced StingrayBench, which evaluates multilingual LLMs on false friends in context across four language pairs, and is the closest prior benchmark to ours in both task design and evaluation setup. The former classifies lexical rela-

tions without context, whereas the latter evaluates usage in context; neither combines both settings, and neither covers Semitic languages.

Despite existing resources and models for Arabic–Hebrew processing, including machine translation, multilingual evaluation, transliteration, and cross-script representation (Shilon et al., 2010; Belinkov and Glass, 2016; Cettolo, 2016; Zaghouani et al., 2024; Gazit et al., 2025; Saeed et al., 2025; Moreno Gonzalez et al., 2026; Gadd, 2026), Arabic–Hebrew cognate relations remain largely unexplored computationally, and to our knowledge, no benchmark exists for evaluating them.

To address this gap, we introduce SemCog Bench, a curated Arabic–Hebrew benchmark of true cognates, false friends, and loanwords for cognate identification and semantic disambiguation.

3 SemCog Bench

In this section, we describe the construction of SemCog Bench. Figure 2 summarizes the benchmark from lexical categories to task formulation. The process consists of four steps: (1) collecting candidate word pairs; (2) word-level annotation; (3) sentence-level validation; and (4) formulating multiple-choice questions for cognate identification and semantic disambiguation.

3.1 Candidate Word Pairs Collection

We collect candidate Arabic–Hebrew word pairs from multiple linguistic resources. For true cognates and false friends, we primarily rely on a comparative Arabic–Hebrew lexicon[4] and the bilingual dictionary of the Palisra project (Habash, 2002, 2023), which contains aligned Arabic and Hebrew word pairs.[5] For loanwords, we compile word candidates from Wiktionary, the Academy of the Hebrew Language materials, and prior studies on loanword phonology, morphological adaptation, and lexical borrowing; the complete list of sources is provided in Appendix A.

Following aggregation, we apply part-of-speech filtering and deduplication, retaining only nouns, verbs, adjectives, and adverbs, as these content words carry the semantic information required for sentence-level interpretation. This yields 1,016 true cognates, 446 false friends, and 724 loanwords. The resulting candidate lists are then reviewed by human annotators for validation.

[4] https://seveleu.com/pages/semitic-syntax-morpho/comparative-sem

[5] https://www.palisra.com/

3.2 Word- and Sentence-Level Annotation

Word-Level Annotation To ensure dataset quality, we employ two fairly compensated annotators with native proficiency in Arabic and Hebrew and formal training in linguistics. One annotator holds an MA in Semitic languages, while the other holds a PhD in linguistics. A primary annotator reviews and accepts or rejects the initial classification of all candidate word pairs. For true cognates and false friends, the annotator assesses the presence of semantic overlap between the Arabic and Hebrew words. Pairs sharing at least one core meaning are retained as true cognates, whereas pairs with completely divergent meanings are retained as false friends. For loanwords, the annotator verifies that both words originate from the same non-Semitic source language and preserve the same meaning. The annotator is also instructed to reject archaic terms, named entities such as person and location names, and lexical items that are not commonly used in both Modern Standard Arabic and Modern Hebrew.

Inter-Annotator Agreement To assess consistency in the word-level annotations, the second annotator independently accepted or rejected the initial classifications for two random 10% subsets: 149 true-cognate/false-friend pairs and 84 loanword pairs. Because both annotators accepted the large majority of the proposed labels, we measure inter-annotator agreement using Gwet's AC1, which is more robust to class imbalance than Cohen's κ and Krippendorff's α (Wongpakaran et al., 2013; Ohyama, 2021). For the true-cognate/false-friend subset, raw agreement was 85%, with a Gwet's AC1 of 0.80. For reference, Cohen's κ and Krippendorff's α were both 0.32 on the same subset. The lower κ and α values are consistent with the skewed marginals, as the primary and second annotators accepted 88% and 86% of the pairs, respectively (Feinstein and Cicchetti, 1990). Of the 23 disagreements (15%), 22 involved cases where one annotator accepted the initial true-cognate or false-friend classification while the other rejected it because they considered the opposite relation more appropriate. The remaining disagreement involved a true-cognate candidate rejected by one annotator as a named entity.

Based on the primary annotator's decisions, we retain 1,858 word pairs: 858 true cognates, 364 false friends, and 636 loanwords as the validated word-level set (Table 1).

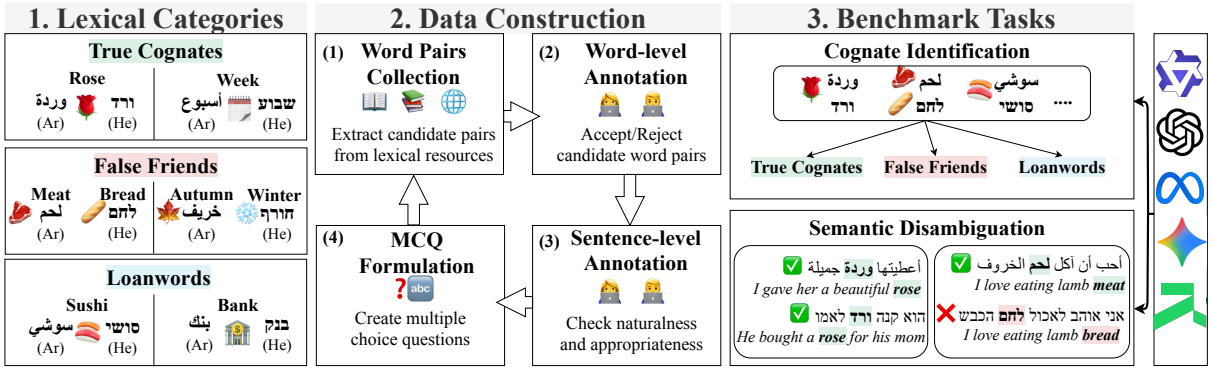


Figure 2: Overview of SemCog Bench, including its lexical categories, data construction, and benchmark tasks.

Sentence-Level Annotation Following word-level validation, we use Gemini-3.1-pro-preview (Team et al., 2025) to generate two contextual sentences for each validated Arabic–Hebrew word pair: one Arabic sentence containing the Arabic target word and one Hebrew sentence containing the corresponding Hebrew target word. This yields 3,716 generated sentences for the 1,858 validated word pairs. The generation prompt is provided in Appendix D.3. One annotator then reviews all generated sentences for naturalness. A sentence is considered natural if it is grammatically correct and representative of native-speaker usage. For sentences deemed unnatural due to grammatical errors or implausible contexts, the annotator provides a revised sentence that preserves the intended meaning and target word. Of the 3,716 generated sentences, 3,549 (9.5%) were retained unchanged, 151 (4.1%) received minor lexical edits, and 16 (0.4%) were substantially rewritten. This review process reduces obvious generation artifacts, including awkward or non-idiomatic phrasing. Detailed annotation guidelines for both the word- and sentence-level annotation tasks are provided in Appendix F.

3.3 Benchmark Tasks

SemCog Bench comprises two benchmark tasks: Cognate Identification (CI) and Semantic Disambiguation (SD). Both tasks are formulated as multiple-choice questions (MCQs) designed to evaluate whether models can distinguish surface-form similarity from semantic equivalence.

Cognate Identification (CI) Given an Arabic–Hebrew word pair (W_{Ar}, W_{He}), the model must determine whether the pair represents a true cognate, false friend, or loanword. This task evaluates whether models can distinguish lexical similarity arising from shared meaning, semantic divergence, or borrowing, rather than relying solely on surface-form resemblance. We consider two input settings. In the *word-only* setting, the model receives only the target word pair. In the *contextual* setting, the model additionally receives contextual sentences in Arabic and Hebrew, (S_{Ar}, S_{He}), providing contextual evidence about the meaning of each word. We evaluate both a three-class setting involving all three categories and a binary setting restricted to true cognates and false friends. The latter removes the confounding effect of loanwords and provides a more direct measure of the model's ability to resolve form–meaning conflicts.

Semantic Disambiguation (SD) Given a word pair (W_{Ar}, W_{He}) and their corresponding contextual sentences (S_{Ar}, S_{He}), the model must determine which sentence or sentences are semantically correct. Each instance is formulated as a three-way MCQ with the following choices: (A) only S_{Ar} is semantically correct; (B) only S_{He} is semantically correct; and (C) both sentences are correct.

For true cognate and loanword pairs, we use the validated sentences collected during sentence-level annotation. Since both sentences employ words with the same meaning, they are semantically valid and the correct answer is therefore (C) (Figure 1(a,d)).

For false friends, we start from the validated Arabic and Hebrew sentences for each false-friend pair and create minimally different sentence pairs in which one sentence is semantically correct and the other is semantically incorrect. For each direction, an annotator translates the validated sentence into the other language and replaces the correct translation of the target word with its false-friend counterpart while preserving agreement and inflection. The resulting sentence remains grammatically

Type	#Pairs	Ar Len.	He Len.	#MCQs
TC	858	13.0	12.8	858
FF	364	13.1	12.8	728
LW	636	14.9	14.4	636
Total	1,858	13.7	13.4	2,222

Table 1: Statistics of SemCog Bench. TC, FF, and LW denote true cognates, false friends, and loanwords, respectively. Ar Len. and He Len. are the average lengths (in words) of the Arabic and Hebrew sentences.

well-formed but becomes semantically implausible (Figure 1(b,c)). We apply this procedure in both directions (Arabic↔Hebrew), yielding two SD instances per false-friend pair.

In total, the SD task contains 2,222 MCQ instances: 858 true cognates, 728 false friends, and 636 loanwords (Table 1).

4 Experimental Setup

We evaluate a diverse set of LLMs on SemCog Bench and examine the effects of input representation, sentence-level context, and model scale.

LLMs We evaluate LLMs categorized into four groups: (1) **Multilingual Models**, including Qwen2.5-7B-Instruct (Qwen et al., 2025), Qwen3-8B (Yang et al., 2025), GLM-4-9B-Chat (GLM et al., 2024), Llama-3.1-8B-Instruct and Llama-3.3-70B-Instruct (Grattafiori et al., 2024), Gemma-2-9B-IT and Gemma-4-31B-IT (Team et al., 2024), Aya-23-8B (Aryabumi et al., 2024), and GPT-OSS-120B (OpenAI et al., 2025); (2) **Arabic-centric Models**, comprising Jais-2-8B-Chat (Anwar et al., 2026) and Jais-Adapted-7B-Chat (Sengupta et al., 2023); (3) **Hebrew-capable Models**, represented by DictaLM-2.0-Instruct (Shmidman et al., 2024); and (4) **Proprietary Models**, which include GPT-5.4 (Singh et al., 2026), GPT-4o (OpenAI et al., 2024), DeepSeek-v4-pro (DeepSeek-AI et al., 2025), and Qwen3.6-Plus (Yang et al., 2025). Proprietary LLM snapshots and licenses for all models are provided in Appendix B, hyperparameters in Appendix C, and prompts in Appendix D.

Baselines We include simple non-generative baselines to contextualize LLM performance. For Cognate Identification (CI), we evaluate LingPy's Sound-Class-Based Phonetic Alignment (SCA) algorithm (List and Moran, 2013), which operates on phonetic transcriptions (IPA), as well as embedding-based approaches following Uban et al. (2025). The latter use aligned FastText embeddings (Grave et al., 2018) and multilingual Sentence-BERT (Reimers and Gurevych, 2019) (`distiluse-base-multilingual-cased-v2`) representations, with cosine distance as the similarity measure. Because these methods distinguish only true cognates from false friends, we evaluate them on the 1,222 true-cognate/false-friend pairs. We do not include analogous baselines for Semantic Disambiguation (SD), as these similarity-based methods cannot assess whether a target word is semantically appropriate in context.

Input Representations To examine the effect of script and phonological form, we evaluate four input representations: the original undiacritized text, diacritized text, IPA transcriptions, and Romanized text. For Arabic, diacritics are obtained using CAMeL Tools (Obeid et al., 2020), and IPA transcriptions are derived from the diacritized forms following the approach of Khalifa et al. (2025) with additional linguistic mapping rules. For Hebrew, *nikkud* and IPA transcriptions are generated using Phonikud (Kolani et al., 2025). Romanized forms for both languages are obtained using Uroman (Hermjakob et al., 2018).

Evaluation Metrics For both CI and SD, we report accuracy as the primary evaluation metric. We additionally report macro-averaged F_1 to account for class imbalance. We also report class-specific accuracy to characterize performance across true cognates, false friends, and loanwords.

5 Results and Analysis

RQ1: Identification and Disambiguation Table 2 presents LLM performance on three-class CI and SD using undiacritized inputs. On CI, Qwen3.6-Plus is the strongest proprietary and overall model, achieving 82.1 accuracy and 77.3 macro F_1, while Gemma-4-31B-IT is the strongest open-source model (78.5 accuracy; 72.5 F_1). On SD, Gemma-4-31B-IT achieves the best overall performance (79.6 accuracy; 79.5 F_1), slightly outperforming the strongest proprietary models, GPT-5.4 and Qwen3.6-Plus, which both reach 79.0 accuracy and 78.9 F_1. Across both tasks, however, performance varies substantially by lexical category.

In CI, most models perform much better on true cognates than false friends; GPT-4o, DictaLM-2.0-Instruct, and Aya-23-8B show the opposite pattern,

Model	Cognate Identification (CI)							Semantic Disambiguation (SD)						
	Acc_{TC}	Acc_{FF}	Acc_{LW}	Acc	P	R	F_1	Acc_{TC}	Acc_{FF}	Acc_{LW}	Acc	P	R	F_1
Proprietary LLMs														
GPT-4o	62.5	71.4	90.7	73.9	73.5	74.9	72.5	**94.6**	35.6	93.1	74.8	75.2	74.2	73.7
GPT-5.4	85.9	47.0	**96.4**	81.9	78.3	76.4	76.9	88.7	57.7	90.2	79.0	79.0	79.1	78.9
DeepSeek-v4-pro	91.6	1.1	83.5	71.1	65.6	58.7	55.8	17.0	50.1	19.3	28.5	25.3	28.3	24.0
Qwen3.6-Plus	87.3	47.8	94.8	**82.1**	**78.6**	**76.6**	**77.3**	91.5	50.8	**94.3**	79.0	79.1	78.9	78.9
Open-source LLMs														
Qwen2.5-7B-Instruct	64.9	64.6	25.0	51.2	63.2	51.5	47.9	43.1	40.7	40.1	41.4	37.1	41.1	36.6
Qwen3-8B	75.2	52.7	18.6	51.4	63.0	48.8	45.1	87.8	6.3	84.1	60.0	61.7	59.5	58.6
Llama-3.1-8B-Instruct	91.7	12.1	0.0	44.7	22.3	34.6	27.1	31.9	30.2	41.3	34.1	39.8	35.1	29.8
Llama-3.3-70B-Instruct	87.9	23.1	92.9	76.9	71.8	68.0	67.5	80.1	8.1	82.7	57.2	57.8	56.9	56.5
GLM-4-9B-Chat	88.0	24.7	29.2	55.5	60.4	47.3	47.1	26.7	52.9	30.3	36.3	23.2	36.1	26.2
Gemma-2-9B-IT	80.7	23.9	66.2	64.6	60.9	56.9	57.1	38.1	13.9	41.2	31.0	42.3	30.9	22.5
Gemma-4-31B-IT	84.0	38.7	93.9	78.5	74.1	72.2	72.5	90.7	**58.0**	89.3	**79.6**	**79.6**	**79.6**	**79.5**
Jais-Adapted-7B-Chat	62.4	2.2	36.9	41.9	41.7	33.8	31.0	33.7	30.6	34.6	32.9	33.8	33.5	18.9
Jais-2-8B-Chat	**96.2**	8.0	0.0	46.0	26.0	34.7	25.2	33.8	29.7	33.3	32.3	13.3	32.9	16.4
DictaLM-2.0-Instruct	33.3	**82.4**	0.2	31.6	48.3	38.6	26.4	34.2	29.5	34.4	32.7	10.9	33.3	16.4
Aya-23-8B	6.3	39.0	90.3	41.4	47.8	45.2	34.7	38.0	26.7	44.6	36.2	55.1	37.0	25.2
GPT-OSS-120B	66.6	54.4	82.7	69.7	70.0	71.2	69.8	47.0	12.2	47.3	35.7	53.7	50.3	46.9

Table 2: Results on three-class cognate identification (CI) and semantic disambiguation (SD) using undiacritized inputs. We report overall and class-specific accuracy (Acc), as well as macro-averaged precision (P), recall (R), and F_1. TC, FF, and LW denote true cognates, false friends, and loanwords, respectively. The best result in each column is shown in **bold**.

Model	Acc_{TC}	Acc_{FF}	Acc	F_1
LingPy	79.1	23.4	62.5	50.9
FastText	61.8	**61.5**	61.7	59.1
SBERT	54.9	**61.5**	56.9	55.0
Qwen3.6-Plus	**94.1**	35.7	**76.7**	66.3
Gemma-4-31B-IT	88.9	43.1	75.3	**68.4**

Table 3: Results on binary cognate identification over 1,222 true-cognate/false-friend (TC/FF) word pairs in terms of accuracy (Acc) and F_1.

while Qwen2.5-7B-Instruct is nearly balanced. Jais-2-8B-Chat provides an extreme example, reaching 96.2 accuracy on true cognates but only 8 on false friends, consistent with a tendency to associate form similarity with shared meaning. Loanwords differ: they are the best-performing category for several models (e.g., GPT-4o, GPT-5.4, Qwen3.6-Plus), yet accuracy falls to 0 for Llama-3.1-8B-Instruct and Jais-2-8B-Chat, highlighting substantial variation in recognizing parallel borrowings.

SD exhibits a more model-dependent class pattern. GPT-4o reverses its CI behavior, shifting from higher false-friend accuracy in CI (71.4 vs. 62.5) to substantially higher true-cognate accuracy in SD (94.6 vs. 35.6). DeepSeek-v4-pro and GLM-4-9B-Chat reverse in the opposite direction, performing better on false friends than true cognates in SD. DictaLM-2.0-Instruct and Aya-23-8B similarly shift from higher false-friend accuracy in CI to higher true-cognate accuracy in SD. Llama-3.1-8B-Instruct also drops sharply on true cognates, from 91.7 in CI to 31.9 in SD, resulting in nearly equal performance on true cognates and false friends. These reversals show that class-specific strengths in CI do not necessarily carry over to SD.

Table 3 compares the simple baselines with Qwen3.6-Plus and Gemma-4-31B-IT, the strongest LLMs, under the same binary CI setting. LingPy performs well on true cognates (79.1) but poorly on false friends (23.4), highlighting the limitations of relying primarily on phonetic similarity. The embedding-based baselines are more balanced across the two classes, with FastText achieving the strongest baseline performance. The LLMs achieve substantially higher overall accuracy, with Qwen3.6-Plus reaching 76.7, while Gemma-4-31B-IT obtains the highest macro F_1 at 68.4. Interestingly, both FastText and SBERT outperform the LLMs on false friends, whereas the LLMs are substantially stronger on true cognates. This suggests that LLMs over-rely on form similarity when distinguishing true cognates from false friends.

	Cognate Identification (CI)						Semantic Disambiguation (SD)					
	Diacritized		IPA		Romanized		Diacritized		IPA		Romanized	
Model	**Acc**	**F_1**	**Acc**	**F_1**	**Acc**	**F_1**	**Acc**	**F_1**	**Acc**	**F_1**	**Acc**	**F_1**
Proprietary LLMs												
GPT-4o	+1.7	+0.6	-15.1*	-13.0*	-32.1*	-30.5*	-29.9*	-29.4*	-45.7*	-46.2*	-45.5*	-47.3*
GPT-5.4	-1.8*	-2.7*	-11.3*	-9.5*	-20.1*	-18.6*	-45.9*	-46.0*	-43.8*	-46.1*	-54.7*	-56.0*
DeepSeek-v4-pro	-3.0*	-3.9*	-10.8*	-11.9*	-17.5*	-19.9*	-9.4*	-7.5*	-10.5*	-6.8*	+3.7*	-6.1*
Qwen3.6-Plus	-2.7	-3.3	-16.0	-15.3	-30.8*	-28.3*	-44.1	-45.0	-42.7	-45.6	-53.8*	-55.1*
Open-source LLMs												
Qwen2.5-7B-Instruct	-2.6*	-5.6*	-25.8*	-25.2*	-21.0*	-20.5*	-8.5*	-4.5*	-12.6*	-5.5*	-20.3*	-17.5*
Qwen3-8B	-3.0*	-7.0*	-22.1*	-20.4*	-27.6*	-26.2*	-8.4*	-12.9*	+3.2*	+3.6*	-30.5*	-37.3*
Llama-3.1-8B-Instruct	+0.9	-1.5*	-0.6	-1.0	-10.0*	+5.3*	-3.0*	-3.8*	+1.9	-5.3*	+12.1*	-6.6*
Llama-3.3-70B-Instruct	-7.1*	-6.0*	-32.2*	-26.9*	-33.0*	-26.3*	-2.0*	-3.2*	-5.4*	-5.3*	-17.8*	-19.2*
GLM-4-9B-Chat	-2.9*	-5.4*	-29.1*	-22.6*	-27.7*	-18.6*	-1.5	-6.0*	-3.0*	-9.6*	-0.9	-18.3*
Gemma-2-9B-IT	-1.1	-5.2*	-24.2*	-26.2*	-16.0*	-21.1*	-1.9*	-5.5*	+1.6*	-6.1*	-10.9*	-16.7*
Gemma-4-31B-IT	-0.7	-2.9*	-11.9*	-9.4*	-21.1*	-18.4*	-43.1*	-42.5*	-46.9*	-46.4*	-54.7*	-54.3*
Jais-Adapted-7B-Chat	+3.8*	-3.7*	-4.1*	-3.3*	-7.0*	-12.8*	-0.2	-2.5*	-0.2	-2.5*	-13.5*	-21.0*
Jais-2-8B-Chat	-0.4	-2.2*	-5.4*	+1.2	-17.2*	+3.1	-0.8*	+0.3	-0.5	+1.0*	+13.2*	+7.8*
DictaLM-2.0-Instruct	+1.2	+5.5*	-12.0*	-15.4*	-12.0*	-15.4*	0.0	0.0	0.0	0.0	0.0	0.0
Aya-23-8B	+0.2	-0.3	-7.8*	-4.0*	-7.5*	-1.4	+3.5*	+7.3*	-3.5*	-8.8*	+16.3*	+10.3*
GPT-OSS-120B	-5.8*	-5.7*	-24.8*	-20.1*	-30.2*	-27.3*	-20.0*	-20.9*	-14.6*	-19.3*	-5.4*	-7.0*

Table 4: Change (Δ) in accuracy (Acc) and F_1 relative to the undiacritized input across different input representations. Cell shading encodes the magnitude and direction of change, from **red** (degradation) to **green** (improvement). Asterisks (*) indicate statistically significant differences ($p < 0.05$).

RQ2: Input Representations Table 4 presents the change in accuracy and F_1 for each input representation relative to the corresponding undiacritized result; significance is assessed using a two-sided approximate randomization test with 10,000 trials (Dror et al., 2018). For CI, diacritization has relatively little effect, producing small gains for some models and modest degradations for others. IPA and Romanization, by contrast, generally reduce performance, often substantially. For example, Llama-3.3-70B-Instruct drops by 32.2 accuracy points and 26.9 F_1 under IPA, and by 33 and 26.3 points, respectively, under Romanization.

For SD, the effect of representation is more model-dependent. The strongest models under undiacritized input often degrade substantially when the representation is changed: GPT-5.4, Qwen3.6-Plus, and Gemma-4-31B-IT lose more than 40 accuracy points in several settings. In contrast, some lower-performing models benefit from particular representations. IPA improves both accuracy and F_1 for Qwen3-8B, while Romanization improves Jais-2-8B-Chat and Aya-23-8B; notably, Aya-23-8B gains 16.3 accuracy points and 10.3 F_1. These results suggest that alternative representations provide no consistent advantage over undiacritized input, with their effects depending strongly on both the task and model.

RQ3: Context Effect on Cognate Identification Table 5 presents the change in accuracy and F_1 from word-level to sentence-level CI across input representations. Among proprietary models, GPT-4o, GPT-5.4, and Qwen3.6-Plus improve across all representations, with larger gains for GPT-4o and GPT-5.4. Several open-source models show consistent gains, including Qwen2.5-7B-Instruct, Llama-3.3-70B-Instruct, and Gemma-4-31B-IT.

Some models benefit particularly strongly from context under alternative input representations. DictaLM-2.0-Instruct gains 19.9 accuracy points and 15.1 F_1 under IPA, while Qwen2.5-7B-Instruct also achieves its largest gains under IPA (+21.3 accuracy; +24.6 F_1). Other models exhibit representation-dependent effects; for example, Gemma-2-9B-IT degrades under undiacritized and diacritized inputs but improves under IPA and Romanization. Aya-23-8B, by contrast, degrades across all representations. Overall, this suggests that the ability to exploit sentence-level context for CI varies substantially across models and interacts strongly with input representation.

Model	Undiacritized		Diacritized		IPA		Romanized	
	Acc	F_1	Acc	F_1	Acc	F_1	Acc	F_1
Proprietary LLMs								
GPT-4o	+15.1*	+15.6*	+14.2*	+15.5*	+18.2*	+16.5*	+17.9*	+17.9*
GPT-5.4	+10.1*	+13.5*	+9.9*	+13.8*	+16.1*	+17.1*	+18.4*	+20.5*
DeepSeek-v4-pro	+2.1*	+2.2*	-9.3*	-2.7*	-10.2*	-4.3*	-1.6	+4.5*
Qwen3.6-Plus	+7.5*	+10.8*	+9.0*	+13.4*	+10.4*	+9.0*	+14.3*	+12.8*
Open-source LLMs								
Qwen2.5-7B-Instruct	+13.0*	+14.2*	+17.4*	+22.0*	+21.3*	+24.6*	+18.4*	+21.3*
Qwen3-8B	+0.6	-3.4*	+1.1	-2.3*	+7.2*	+5.5*	+11.4*	+5.2*
Llama-3.1-8B-Instruct	+2.0*	-4.7*	-1.1*	-4.2*	-34.0*	-16.3*	-3.7*	-5.0*
Llama-3.3-70B-Instruct	+5.1*	+7.1*	+7.1*	+4.3*	+11.1*	+7.7*	+16.5*	+7.3*
GLM-4-9B-Chat	-2.4	-0.3	-2.6	+5.5*	+3.7*	+5.8*	-3.4*	-4.0*
Gemma-2-9B-IT	-7.1*	-10.4*	-3.2*	-5.2*	+4.4*	+4.3*	+3.9*	+1.5
Gemma-4-31B-IT	+12.5*	+17.4*	+12.3*	+19.0*	+13.3*	+15.0*	+12.2*	+14.8*
Jais-Adapted-7B-Chat	+5.1*	-1.9	-1.1	-4.8*	-3.7*	-10.6*	+7.5*	+9.7*
Jais-2-8B-Chat	-5.5*	+5.8*	-5.0*	+9.0*	-5.0*	-0.8	-6.3*	-5.3*
DictaLM-2.0-Instruct	+11.1*	+3.4*	+12.8*	-7.7*	+19.9*	+15.1*	+6.4*	+9.5*
Aya-23-8B	-13.1*	-11.6*	-6.8*	-2.4	-12.9*	-16.8*	-6.1*	-0.2
GPT-OSS-120B	+11.2*	+12.0*	+8.6*	+9.0*	+0.2	+2.4	+5.5*	+7.3*

Table 5: Change (Δ) in accuracy (Acc) and F_1 from word-level to sentence-level cognate identification in the three-class setting. Cell shading indicates the direction and magnitude of change, ranging from **red** (degradation) to **green** (improvement). Asterisks (*) denote statistically significant differences ($p < 0.05$).

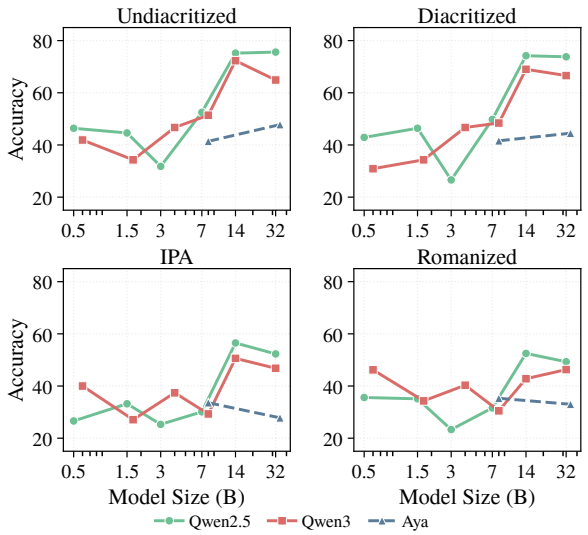


Figure 3: Effect of model scale on cognate identification accuracy across different input representations for the Qwen2.5, Qwen3, and Aya-23 model families.

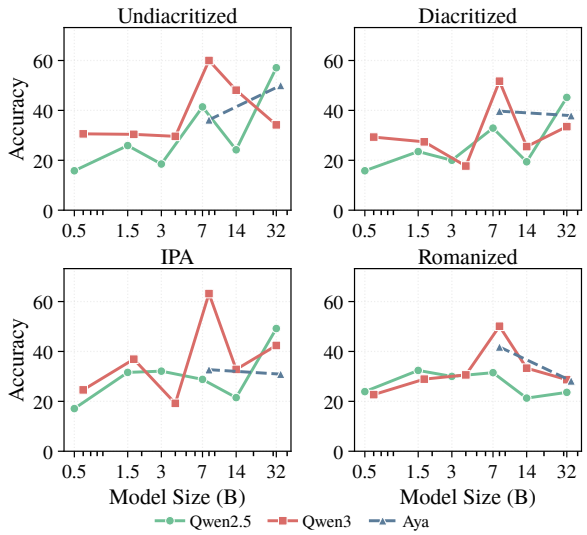


Figure 4: Effect of model scale on semantic disambiguation accuracy across different input representations for the Qwen2.5, Qwen3, and Aya-23 model families.

RQ4: Impact of Model Size We compare multiple scales of Qwen2.5, Qwen3, and Aya-23, covering sub-billion to 35B-parameter models. Figure 3 shows that scaling generally improves CI, but gains vary across model families and representations. For Qwen2.5 and Qwen3, performance improves substantially up to 14B under undiacritized and diacritized inputs, while scaling to 32B provides little additional benefit or slightly degrades performance. Gains are smaller under IPA and Romanization, where accuracy remains substantially lower. Aya shows modest gains with scale under undiacritized input, but little or no improvement under alternative representations.

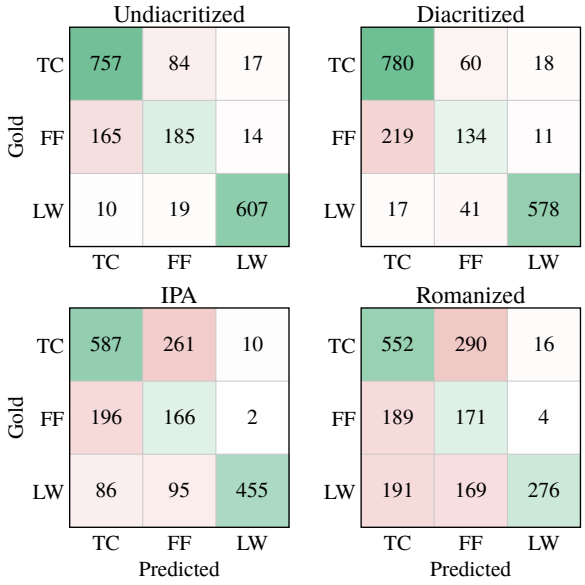


Figure 5: Qwen3.6-Plus confusion matrices for cognate identification across input representations. TC, FF, and LW denote true cognates, false friends, and loanwords.

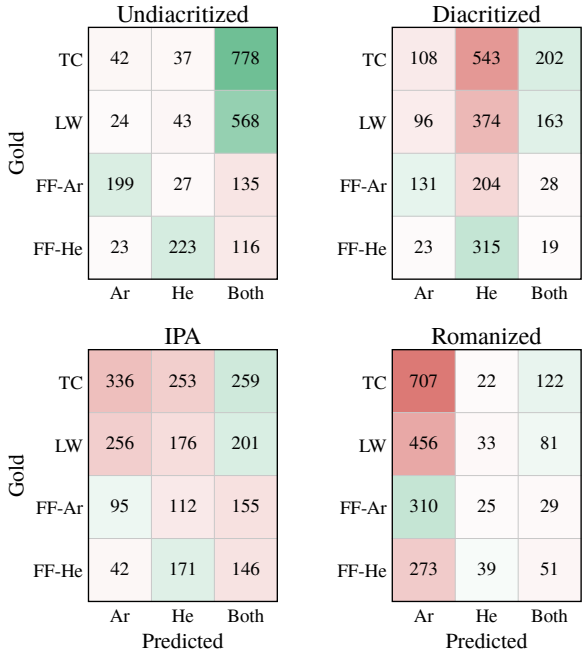


Figure 6: Gemma-4-31B-IT confusion matrices for semantic disambiguation across input representations. TC/LW denote true cognates/loanwords; FF-Ar/FF-He denote whether only the Arabic/Hebrew sentence is correct.

SD exhibits a less consistent scaling pattern as shown in Figure 4. Qwen2.5 generally improves with scale, with large gains from 14B to 32B, whereas Qwen3 peaks at 8B across all representations and drops sharply at 14B. Aya again shows only modest changes across sizes. Scaling therefore benefits CI more under undiacritized inputs, while gains on SD are less predictable and depend strongly on model family and input representation.

5.1 Error Analysis

To better understand errors in the strongest LLMs, Figures 5 and 6 show confusion matrices for Qwen3.6-Plus, the best LLM on CI, and Gemma-4-31B-IT, the best on SD, across four input representations. In CI, errors are dominated by confusion between true cognates and false friends: true cognates are usually predicted as false friends, while false friends are overwhelmingly predicted as true cognates. Loanword errors differ: under undiacritized, diacritized, and IPA inputs, they are more often confused with false friends, whereas Romanization shifts them toward true cognates, further suggesting reliance on surface-form similarity.

In SD, representation strongly affects which sentence the model favors. For true cognates and loanwords, where the correct answer is *Both*, undiacritized input yields high accuracy; diacritization favors Hebrew, while IPA and Romanization shift errors toward Arabic. For false friends, undiacritized errors are mostly *Both*. Diacritization improves Hebrew-only cases (223 to 315 correct) but shifts 204 Arabic-only cases to Hebrew-only. Under IPA, *Both* again dominates, whereas Romanization favors Arabic, with 310 correct Arabic-only predictions and 273 Hebrew-only cases shifted to Arabic-only. Appendix E presents the same error analysis across CI and SD for each model.

6 Conclusion and Future Work

We introduced SemCog Bench, a manually curated benchmark of 1,858 Arabic–Hebrew true cognates, false friends, and loanwords with word- and sentence-level annotations for cognate identification and semantic disambiguation. Across a diverse set of LLMs, we find strong reliance on surface-form similarity, with better performance on true cognates than false friends and wide variation on loanwords. Input representation, sentence-level context, and model scale all affect performance, but their effects vary across models and tasks. Original-script input is generally strongest, while context and scaling are more model-dependent.

In future work, we plan to extend SemCog Bench to additional language pairs and develop methods that better integrate orthographic, phonological, and contextual signals for cross-lingual lexical reasoning.

Limitations

SemCog Bench focuses on Arabic and Hebrew, so the findings may not generalize to language pairs with different typological, script, or contact patterns. Our evaluation uses controlled multiple-choice tasks, which enable systematic comparison but may not fully reflect open-ended settings such as generation, translation, or dialogue. The sentence contexts were initially generated with Gemini and subsequently reviewed and revised by human annotators; nevertheless, residual lexical or syntactic patterns from the generation process may remain. Finally, some input representations rely on automatic diacritization, Romanization, and IPA transcription tools, and proprietary-model results may vary with future model or API updates despite our use of fixed snapshots.

Ethics Statement

This work uses linguistic resources and human-validated sentence examples to study Arabic–Hebrew lexical reasoning in LLMs. The dataset does not contain personal, private, or sensitive user information. Annotators have relevant linguistic expertise, and the annotation process includes independent verification of a subset of word-level labels. A key consideration is to avoid over-interpreting model behavior as reflecting properties of the languages or their speakers; our analysis is intended to diagnose limitations of models, not to evaluate linguistic communities. By identifying errors involving false friends and loanwords, this work aims to support more reliable multilingual systems for applications such as translation, language learning, and cross-lingual information access. We used AI writing assistance within the scope of "Assistance purely with the language of the paper" described in the ACL Policy on Publication Ethics.

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A Data Resources

We compiled candidate true cognates, false friends, and loanwords from a variety of dictionaries, linguistic databases, and academic publications.

True Cognates and False Friends Candidate true cognate and false friend pairs were collected from:

- A Comparative Arabic–Hebrew Lexicon [Source].
- The Palisri Dictionary of the Palisra Project [Source].

Loanwords Candidate loanwords were collected from:

- Hebrew loanword resources, including educational materials, Academy of the Hebrew Language documentation, and research on Modern Hebrew loanword phonology [Source 1, Source 2, Source 3, Source 4].
- Arabic loanword resources, including studies of English loanwords, morphological adaptation, and lexical borrowing in Modern Standard Arabic [Source 1, Source 2, Source 3, Source 4].

B LLM Snapshots and Licenses

Snapshots To ensure the reproducibility and consistency of our experimental results, all evaluations involving proprietary models were executed using fixed, version-controlled API snapshots. Specifically, we used OpenAI's `gpt-5.4-2026-03-05` and `gpt-4o-2024-11-20`, DeepSeek's `deepseek-v4-pro`, and Alibaba Cloud's `qwen3.6-plus-2026-04-02`.

Licenses

- **Apache 2.0:** Qwen2.5-7B-Instruct (Qwen et al., 2025), Qwen3-8B (Yang et al., 2025), Gemma-4-31B-IT (Team et al., 2024), GPT-OSS-120B (OpenAI et al., 2025), Jais-2-8B-Chat (Anwar et al., 2026), Jais-Adapted-7B-Chat (Sengupta et al., 2023), and DictaLM-2.0-Instruct (Shmidman et al., 2024).
- **Llama Community License:** Llama-3.1-8B-Instruct and Llama-3.3-70B-Instruct (Grattafiori et al., 2024).
- **Gemma Terms of Use:** Gemma-2-9B-IT (Team et al., 2024).
- **CC-BY-NC-4.0:** Aya-23-8B (Aryabumi et al., 2024).
- **GLM-4 License:** GLM-4-9B-Chat (GLM et al., 2024).
- **MIT:** DeepSeek-v4-pro (DeepSeek-AI et al., 2025).
- **OpenAI Services Agreement:** GPT-5.4 (Singh et al., 2026) and GPT-4o (OpenAI et al., 2024).
- **Alibaba Cloud Model Studio Terms:** Qwen3.6-Plus (Yang et al., 2025).

C Compute and Hyperparameters

We evaluate the open-source models on two NVIDIA A40 GPUs, each with 48 GB of memory, and access the proprietary models through their official APIs. All experiments with open-source models are conducted using the LM Evaluation Harness framework[6], which provides a consistent evaluation interface and supports reproducible comparisons.

For open-source model inference, we use the vLLM framework (Kwon et al., 2023) to improve computational efficiency and memory usage. To ensure reproducibility, we use deterministic decoding with temperature $t = 0$ and limit the maximum output length to $max_token = 20$. For proprietary models, including GPT, DeepSeek, and Qwen, we use the same decoding configuration for t and max_token whenever the corresponding API supports these parameters. We exclude the Gemini model family from the evaluation to avoid potential circular evaluation bias. To reduce position bias in the semantic disambiguation task, we randomly permute the order of answer choices and construct a fixed evaluation set with randomized option labels. All results reported in this work are computed on this fixed evaluation set.

D LLMs Prompts

D.1 Cognate Identification

Cognate Identification (Undiacritized)

```
You are provided with an Arabic word and a
Hebrew word. Your task is to classify the
relationship between the words into one of
the following categories: - TRUE_COGNATE -
FALSE_FRIEND - LOANWORD
Provide only one label as the response:
```

[6] https://github.com/EleutherAI/lm-evaluation-harness

TRUE_COGNATE, FALSE_FRIEND, or LOANWORD.
Definitions:
TRUE_COGNATE: Words that share a common Semitic root and possess semantically overlapping meanings. At least one core meaning must be present in both languages.
FALSE_FRIEND: Words that exhibit similar orthographic or phonological forms but convey completely different meanings, with no meaningful semantic overlap.
LOANWORD: Words that are borrowed from one language into another, resulting in similar forms without a shared Semitic root.
Arabic: {arabic_word}
Hebrew: {hebrew_word}
Answer:

Cognate Identification (Diacritized)

You are provided with a fully diacritized Arabic word and a fully diacritized Hebrew word. Your task is to classify the relationship between the words into one of the following categories: - TRUE_COGNATE - FALSE_FRIEND - LOANWORD
Provide only one label as the response: TRUE_COGNATE, FALSE_FRIEND, or LOANWORD.
Definitions:
TRUE_COGNATE: Words that share a common Semitic root and possess semantically overlapping meanings. At least one core meaning must be present in both languages.
FALSE_FRIEND: Words that exhibit similar diacritized orthographic or phonological forms (including vowel patterns and consonantal skeletons) but convey completely different meanings, with no meaningful semantic overlap.
LOANWORD: Words that are borrowed from one language into another, resulting in similar diacritized forms without a shared Semitic root.
Arabic (diac): {arabic_word_diac}
Hebrew (diac): {hebrew_word_diac}
Answer:

Cognate Identification (IPA)

You are provided with an IPA transcription of an Arabic word and a Hebrew word. Your task is to classify the relationship between the words into one of the following categories: - TRUE_COGNATE - FALSE_FRIEND - LOANWORD
Provide only one label as the response: TRUE_COGNATE, FALSE_FRIEND, or LOANWORD.
Definitions:
TRUE_COGNATE: Words that share a common Semitic root and possess semantically overlapping meanings. At least one core meaning must be present in both languages.
FALSE_FRIEND: Words that exhibit similar phonetic forms in IPA transcription (reflecting similar pronunciation) but convey completely different meanings, with no meaningful semantic overlap.
LOANWORD: Words that are borrowed from one language into another, resulting in similar IPA transcriptions without a shared Semitic root.
Arabic (IPA): {arabic_word_ipa}
Hebrew (IPA): {hebrew_word_ipa}
Answer:

Cognate Identification (Romanized)

You are provided with a Universal Romanization of an Arabic word and a Hebrew word. Your task is to classify the relationship between the words into one of the following categories: - TRUE_COGNATE - FALSE_FRIEND - LOANWORD
Provide only one label as the response: TRUE_COGNATE, FALSE_FRIEND, or LOANWORD.
Definitions:
TRUE_COGNATE: Words that share a common Semitic root and possess semantically overlapping meanings. At least one core meaning must be present in both languages.
FALSE_FRIEND: Words that exhibit similar romanized forms (reflecting similar phonological or orthographic structure) but convey completely different meanings, with no meaningful semantic overlap.
LOANWORD: Words that are borrowed from one language into another, resulting in similar romanized forms without a shared Semitic root.
Arabic (uroman): {arabic_word_uroman}
Hebrew (uroman): {hebrew_word_uroman}
Answer:

Cognate Identification (Undiacritized with Context)

You are given an Arabic sentence and a Hebrew sentence, along with a specific target Arabic word and a target Hebrew word extracted from these sentences.
Your task is to determine the relationship between the target Arabic word and the target Hebrew word, taking into account their context within the sentences.
Determine whether they are: - TRUE_COGNATE - FALSE_FRIEND - LOANWORD
Answer with only one label: TRUE_COGNATE, FALSE_FRIEND, or LOANWORD.
Definitions:
TRUE_COGNATE: Words that share a common Semitic root AND have semantically overlapping meanings in the given context. At least one core meaning must be present in both languages.
FALSE_FRIEND: Words that have similar orthographic or phonological forms BUT have completely different meanings in the given context, with no meaningful semantic overlap.
LOANWORD: Words that are borrowed from one language into another, resulting in similar forms without a shared Semitic root.
Arabic Sentence: {arabic_sentence} Target

```
Arabic Word: {arabic_word}
Hebrew Sentence: {hebrew_sentence} Target
Hebrew Word: {hebrew_word}
Answer:
```

D.2 Semantic Disambiguation

Semantic Disambiguation (Undiacritized)

```
You are provided with two sentences: one written in the Arabic script and one in the Hebrew script. Which sentence is semantically appropriate?
A. {sent_a}
B. {sent_b}
C. Both sentences are appropriate.
Provide only the corresponding letter (A, B, or C) as the response.
Answer:
```

Semantic Disambiguation (Diacritized)

```
You are provided with two fully diacritized sentences: one in Arabic script and one in Hebrew script. Which sentence is semantically appropriate?
A. {sent_a}
B. {sent_b}
C. Both sentences are appropriate.
Provide only the corresponding letter (A, B, or C) as the response.
Answer:
```

Semantic Disambiguation (IPA)

```
You are provided with two sentences in IPA transcription: one from Arabic and one from Hebrew. Which sentence is semantically appropriate?
A. {sent_a}
B. {sent_b}
C. Both sentences are appropriate.
Provide only the corresponding letter (A, B, or C) as the response.
Answer:
```

Semantic Disambiguation (Romanized)

```
You are provided with two sentences in transliteration by uroman: one from Arabic and one from Hebrew. Which sentence is semantically appropriate?
A. {sent_a}
B. {sent_b}
C. Both sentences are appropriate.
Provide only the corresponding letter (A, B, or C) as the response.
Answer:
```

D.3 Sentence Generation

Sentence Generation

```
You are provided with an Arabic word, a Hebrew word, their corresponding meanings, and a cognate type. Your task is to generate one natural Arabic sentence and one natural Hebrew sentence, each containing the corresponding target word.
Inputs:
Arabic word: {arabic_word}
Hebrew word: {hebrew_word}
Arabic meaning: {arabic_meaning}
Hebrew meaning: {hebrew_meaning}
Cognate type: {cognate_type}
Output format:
Return only valid JSON with the following fields: arabic_word, hebrew_word, arabic_sentence, hebrew_sentence, arabic_translation_en, and hebrew_translation_en.
Sentence requirements:
Each sentence must contain 8–20 words.
Each sentence must be natural, fluent, and representative of native-speaker usage.
The target word must be clearly used in its given meaning. Inflection is allowed.
Cognate constraints:
If cognate_type is true_cognate, both target words should be used with the same meaning, but the Arabic and Hebrew sentences should describe different situations.
If cognate_type is false_friend, the Arabic sentence must reflect the Arabic meaning, and the Hebrew sentence must reflect the Hebrew meaning. The difference between the two meanings must be clear from context.
Answer:
```

E Per-Model Error Analysis

E.1 Cognate Identification

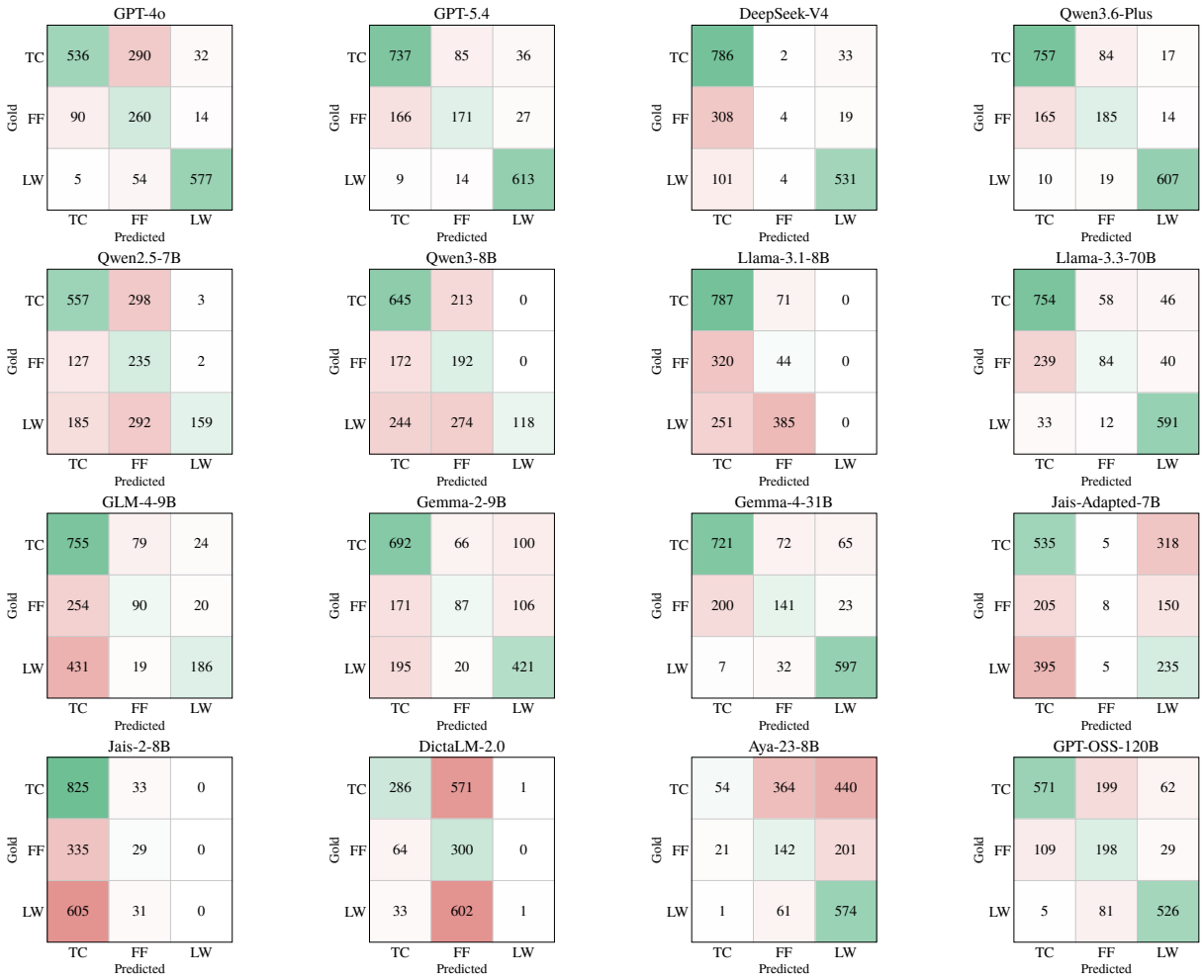


Figure 7: Per-model confusion matrices for cognate identification using original undiacritized input. TC, FF, and LW denote true cognates, false friends, and loanwords.

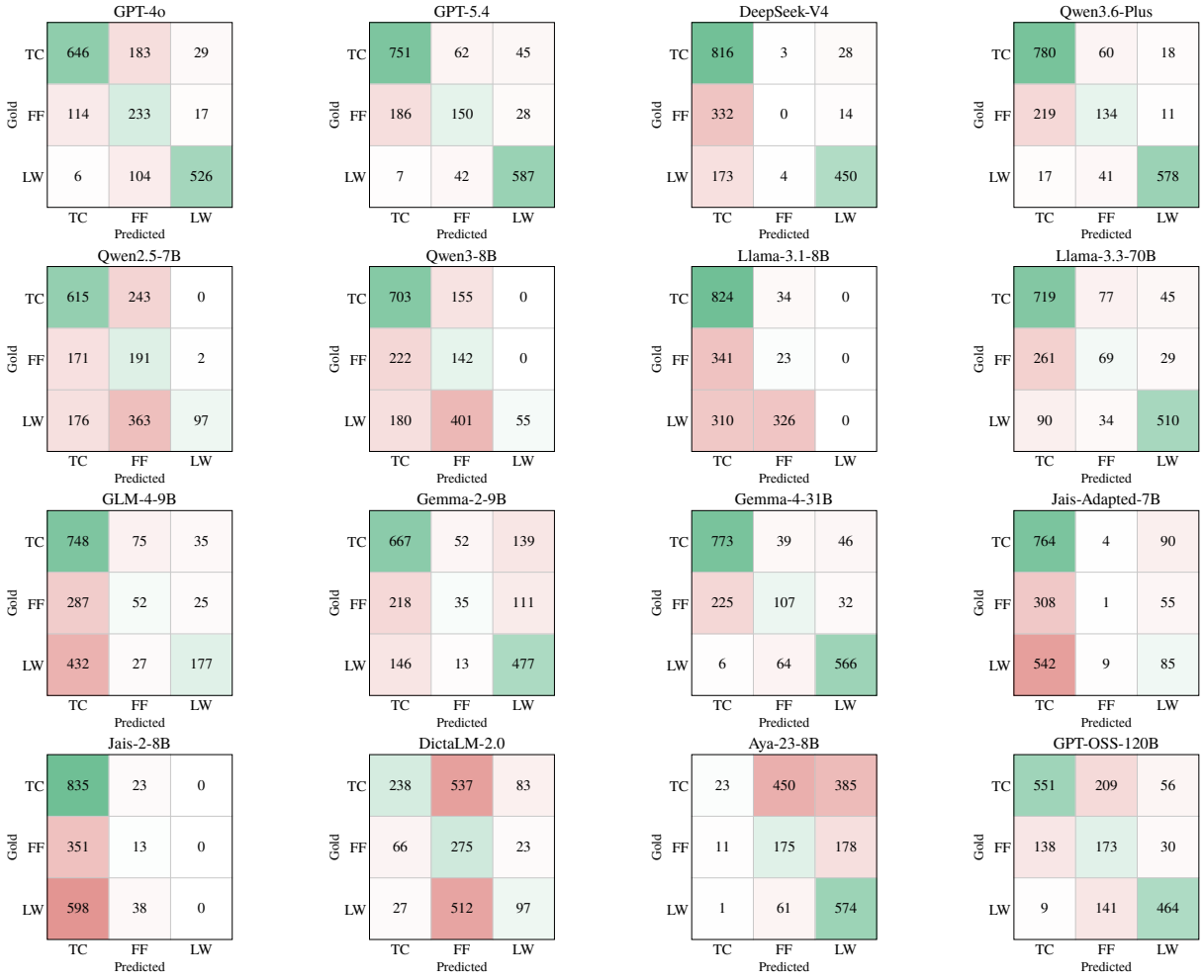


Figure 8: Per-model confusion matrices for cognate identification using diacritized input. TC, FF, and LW denote true cognates, false friends, and loanwords.

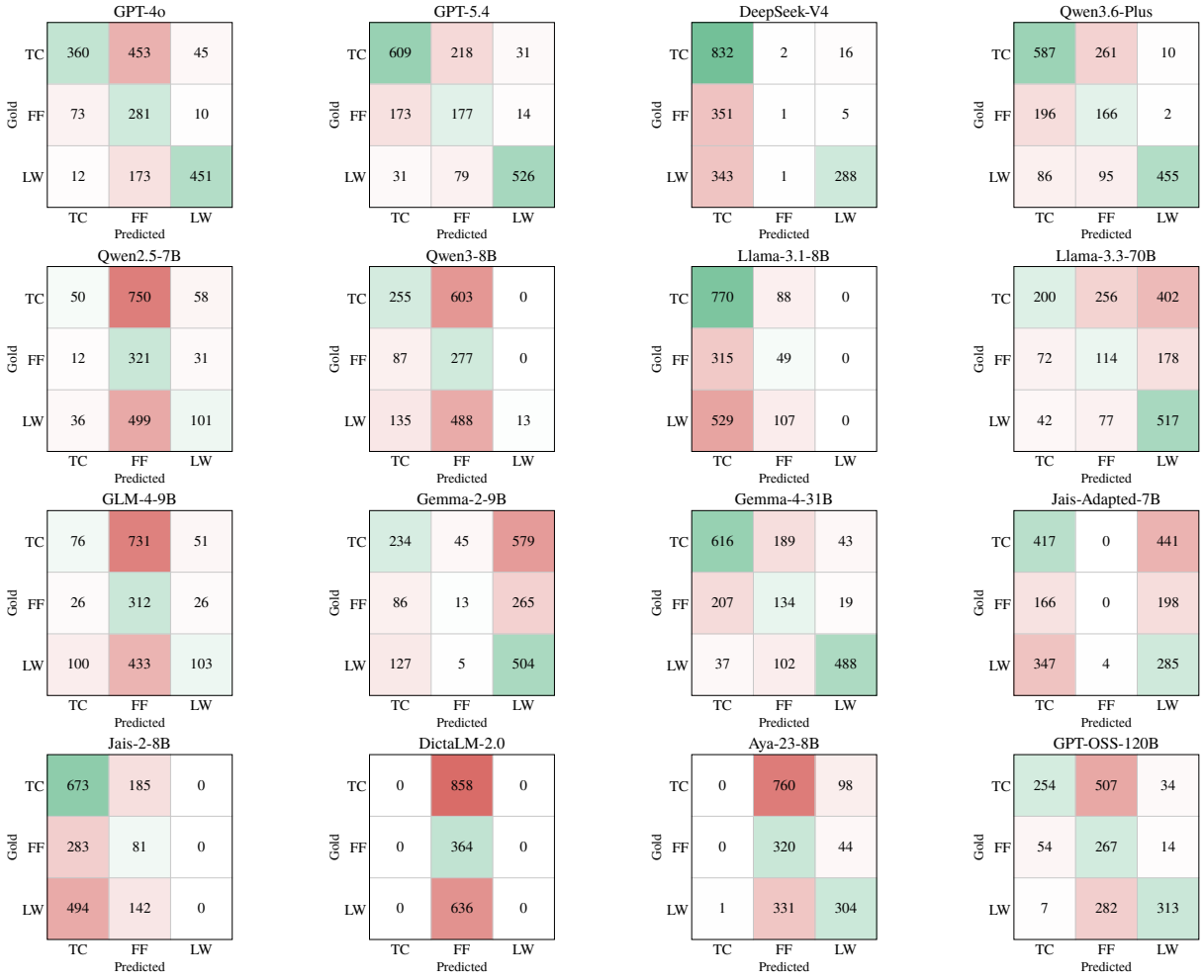


Figure 9: Per-model confusion matrices for cognate identification using IPA input. TC, FF, and LW denote true cognates, false friends, and loanwords.

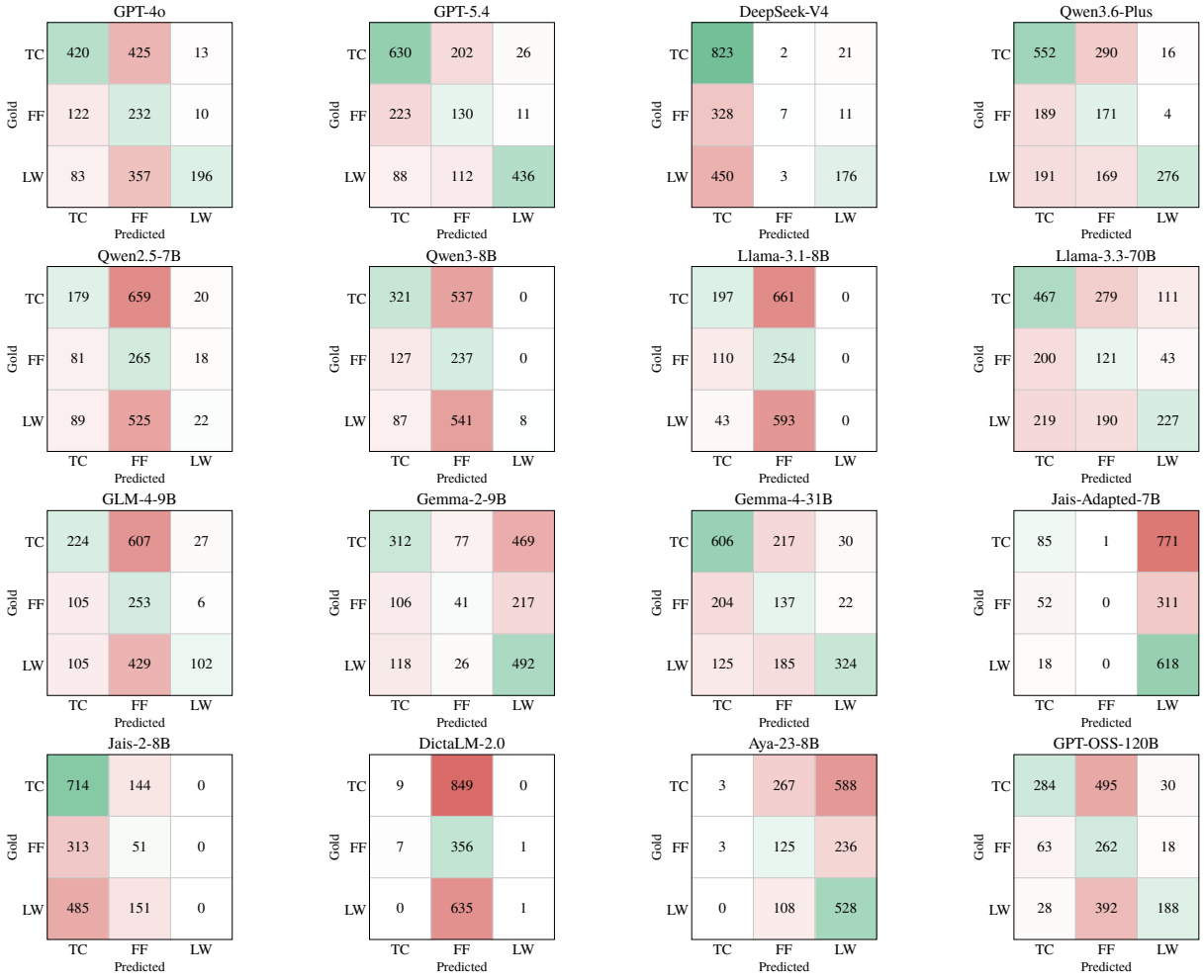


Figure 10: Per-model confusion matrices for cognate identification using Romanized input. TC, FF, and LW denote true cognates, false friends, and loanwords.

E.2 Semantic Disambiguation

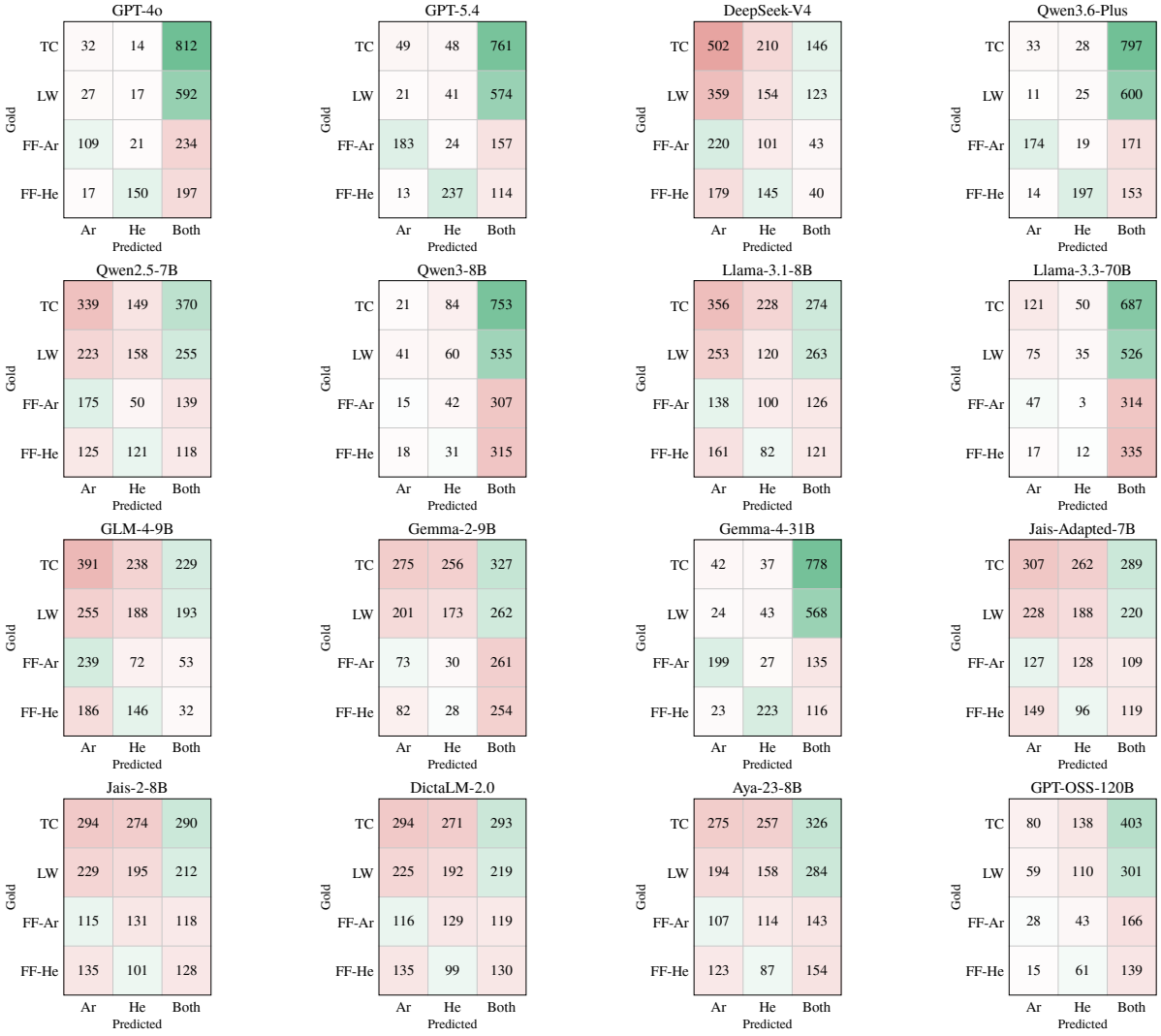


Figure 11: Per-model confusion matrices for semantic disambiguation with undiacritized input. TC/LW denote true cognates/loanwords; FF-Ar and FF-He indicate whether only the Arabic or Hebrew sentence is correct.

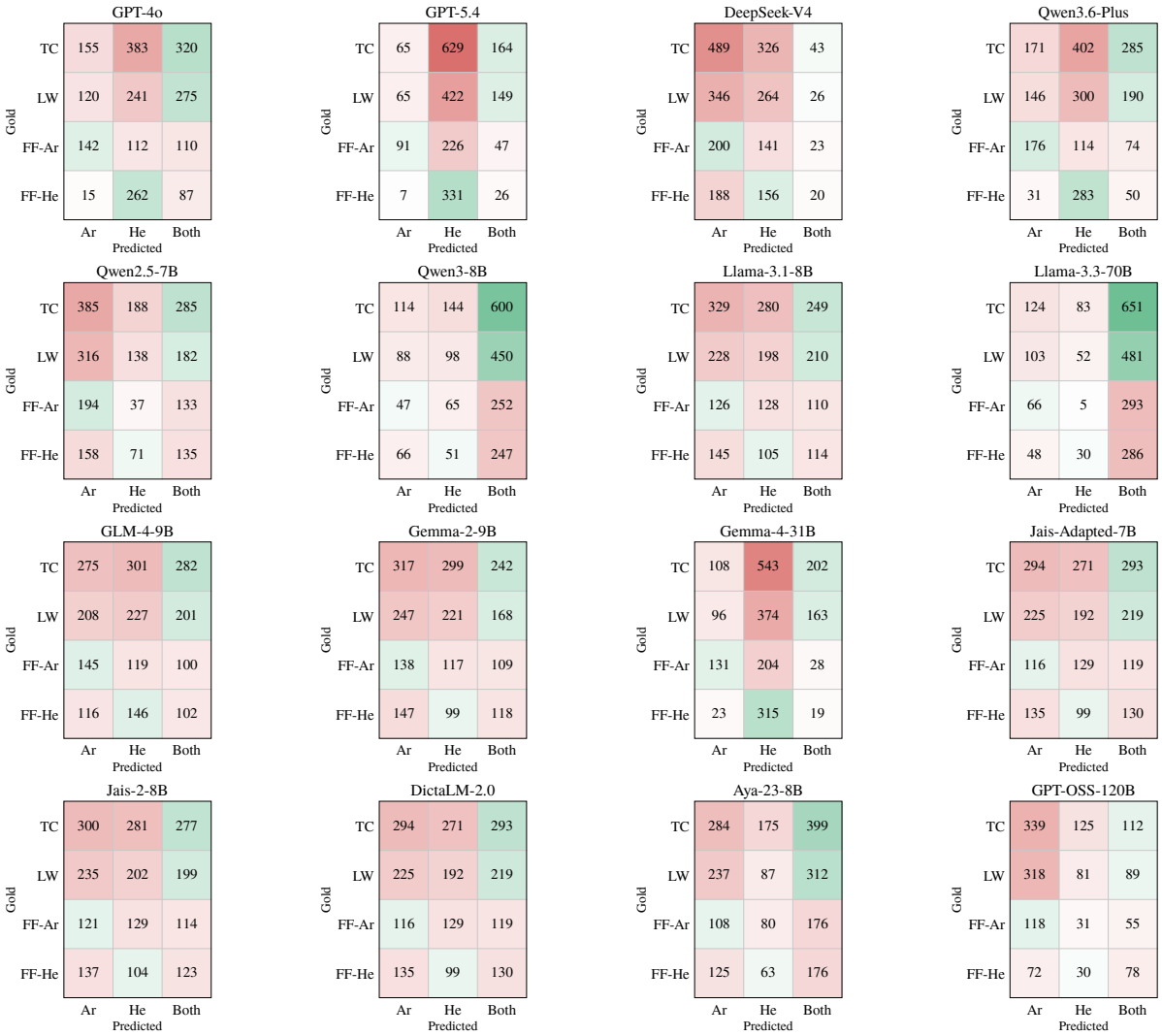


Figure 12: Per-model confusion matrices for semantic disambiguation with diacritized input. TC/LW denote true cognates/loanwords; FF-Ar and FF-He indicate whether only the Arabic or Hebrew sentence is correct.

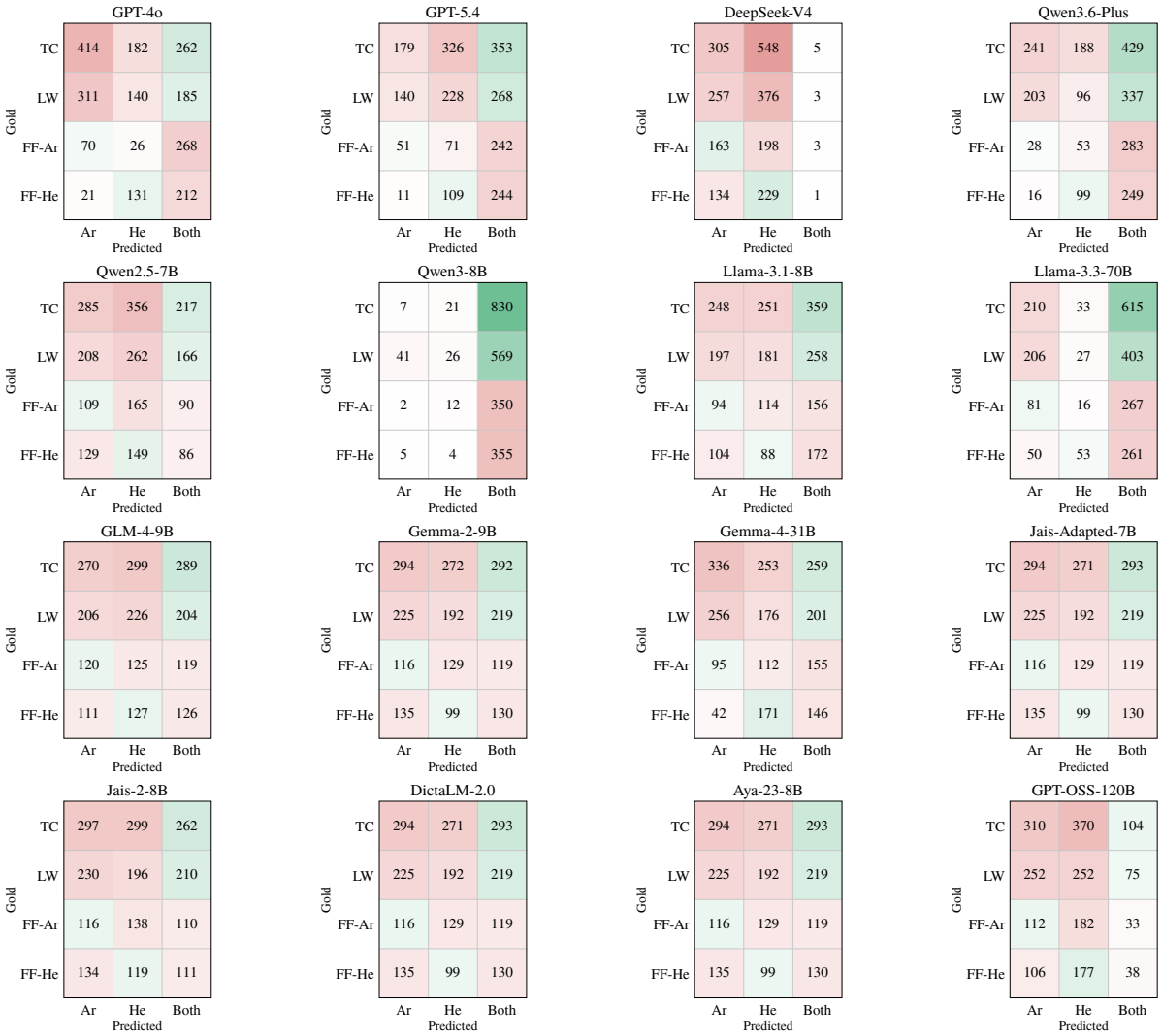


Figure 13: Per-model confusion matrices for semantic disambiguation with IPA input. TC/LW denote true cognates/loanwords; FF-Ar and FF-He indicate whether only the Arabic or Hebrew sentence is correct.

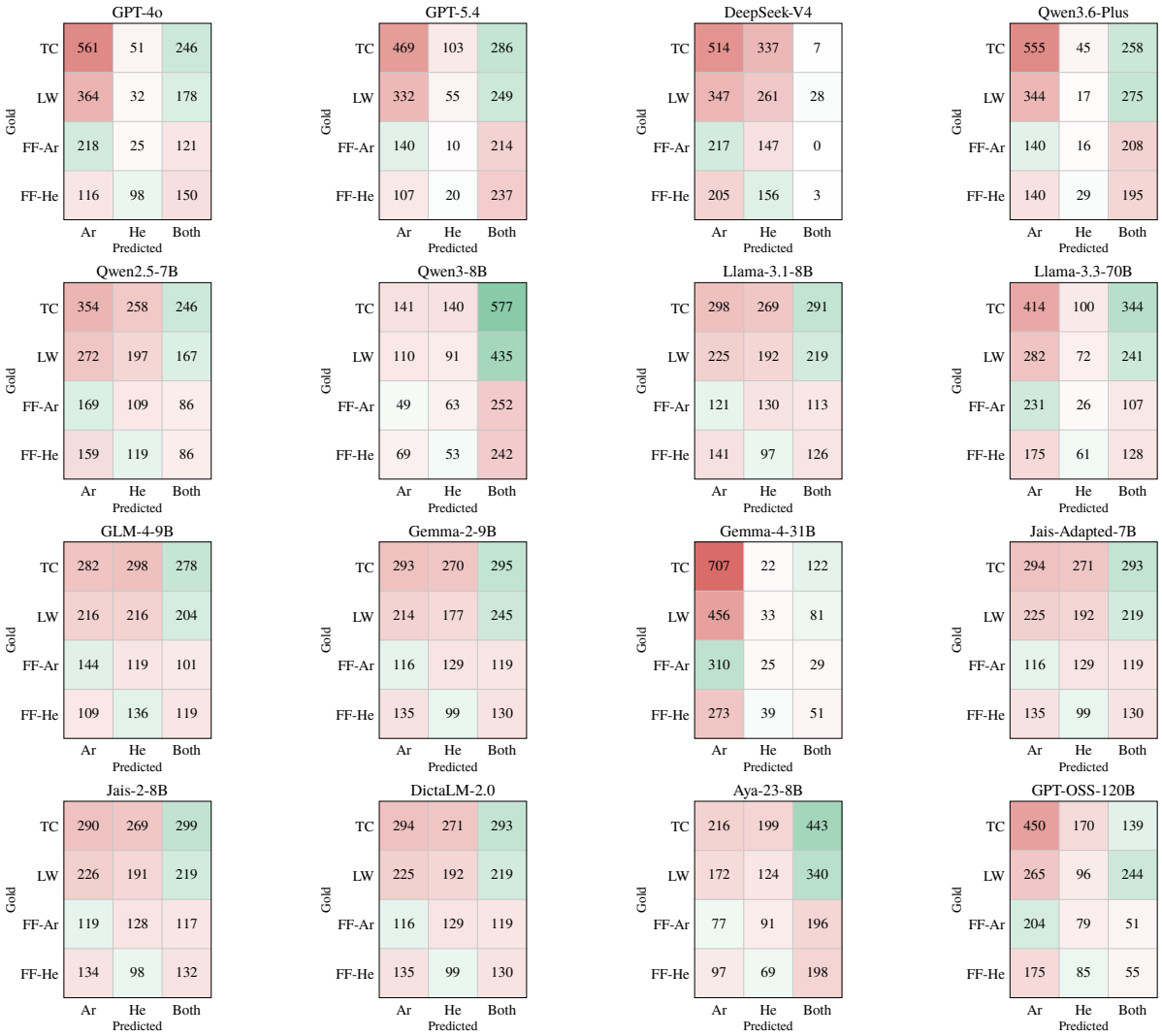


Figure 14: Per-model confusion matrices for semantic disambiguation with Romanized input. TC/LW denote true cognates/loanwords; FF-Ar and FF-He indicate whether only the Arabic or Hebrew sentence is correct.

F Annotation Guidelines

Annotation Guideline: Arabic-Hebrew Cognates Classification

1. Task Overview

The objective of this annotation task is to classify Arabic-Hebrew word pairs as either **True Cognates** or **False Friends** based on semantic equivalence. These words were initially classified using LLMs. Your role as an annotator to verify this classification.

After that, you will be required to classify whether Arabic-Hebrew word pairs are **loanwords**.

Your task will be:

- Decide based on the meaning of the word and decide whether the pair are True Cognates or False Friends (Stage 1)
- Evaluate sentence for naturalness for both Arabic and Hebrew (Stage 2)
- **Decide what is the meaning of the word and decide whether the pair are loanwords (Stage 3)**
- **Evaluate sentence for naturalness for both Arabic and Hebrew for loanwords (Stage 4)**

We will provide you with an annotation spreadsheet. Below are the entries in the annotation spreadsheet.

Stage 1 (Word-Level) - provided in tab 1:

Field	Description
arabic_diac	Arabic word with diacritics
arabic_undiac	Arabic word without diacritics
hebrew_diac	Hebrew word with diacritics
hebrew_undiac	Hebrew word without diacritics
arabic_meaning	English meaning(s) of the Arabic word
hebrew_meaning	English meaning(s) of the Hebrew word

Stage 2 (Sentence-Level) (provided in tab 2):

Field	Description

arabic_word	Arabic target word
hebrew_word	Hebrew target word
arabic_sentence	Sentence in Arabic using the target word
hebrew_sentence	Sentence in Hebrew using the target word
arabic_translation_en	English translation of the Arabic sentence
hebrew_translation_en	English translation of the Hebrew sentence

Stage 3 (Loanwords, word-level annotation)

Field	Description
arabic_word	Arabic loanword
hebrew_word	Hebrew loanword
arabic_meaning	English meaning of Arabic word
hebrew_meaning	English meaning of Arabic word

Stage 4 (Loanwords, sentence-level annotation)

Same as stage 2.

2. Annotation Criteria

2.1 Stage 1: Cognate Annotation

Label	Definition
True Cognate	Words which share a common Semitic root AND have semantically overlapping meanings (share a meaning). At least one core meaning must be present in both languages. (ex. The words חלב and حليب which mean milk are true cognates)

False Friend	Words share similar form BUT have completely different meanings with no meaningful overlap. (ex. خريف and חורף are false friends)

How do decide on the correct meaning:

1. Read the two words and their meanings, is the meaning correct?
2. Check if any meaning overlaps between the two words from the two languages
3. If they have one shared meaning, then the words are **True Cognate**
4. If you accept the classification in column H (type), be it True cognate or False friends, then in annotator_type (column K), choose **Accept**, otherwise, choose **Reject**.
5. Please **Reject** the borrowed word from English, archaic words (i.e. words that are not commonly used in the two languages), only keep words that are used in MSA and Modern Hebrew. Also reject, and names of places/locations, animals etc. As they are not considered cognates.
6. If you have any comments, add a note in column L.
7. *You will also see a "reasoning" column; this is generated by Gemini-3.1-pro-preview. This is only for researchers' references;* ***please*** *use your own judgement when providing your annotation!*

Google Sheet Columns to Fill:

Column	Required	Description
annotator_type (col. K)	Yes	Accept/Reject
note	Optional	Brief explanation or comment (if needed)

A	B	C	D	E	F	G	H	K	L
entry_id	arabic_diac	arabic_un	hebrew_diac	hebrew_undiac	arabic_meaning	hebrew_meanin	type	annotator_type	note
1	عزر	عزر	עזרה	עזרה	reprimand;censu	to help, to assist	False Friend		
2	عود	عود	עוד	עוד	lute;stick; lute;sti	yet, still; more (f	False Friend		
3	أنْ	ان	אֲנָה	אנה	if;whether; that;	to cause, to indu	False Friend		

2.2 Stage 2: Sentence Naturalness

In the second stage you'll decide on the sentence's naturalness. In this task, there are two labels: 1. Natural 2. Unnatural

Label	Definition

Natural	Grammatically correct and idiomatic; for the Hebrew, the sentence should sound like a native speaker production, and for the MSA it sounds like a natural choice in MSA – i.e. no awkwardness in the choice of context.
Unnatural	Contains grammatical errors, awkward phrasing, or would not be produced by a native speaker (for Hebrew).

Google Sheet Columns to Fill:

Column	Required	Description
arabic_sentence_natural	Yes	Enter: "Natural" or "Unnatural"
hebrew_sentence_natural	Yes	Enter: "Natural" or "Unnatural"
arabic_note	Optional	Comment on Arabic sentence (if Unnatural or needs clarification)
hebrew_note	Optional	Comment on Hebrew sentence (if Unnatural or needs clarification)

	A	B	C	D	E	F	G	H	I	J
1	entry_id	arabic_word	hebrew_word	type	arabic_sentence	hebrew_sentenc	arabic_translatic	hebrew_translat	arabic…	hebre…
2	1	عزر	עזרה	False Friend	התלמיד ביקש ע كان عزر المدير للموة		The manager's r	The student aske		
3	2	عود	עוד	False Friend	אני צריך עוד זמן عزف الموسيقي بمها		The musician pla	I need more time		
4	3	أن	אנה	False Friend	הגורל אנה לו אי قرر المعلم أن يؤجل ا		The teacher deci	Fate brought up		
5	4	استعبد	שעבד	True Cognate	הכובש הזר שעב على مر التاريخ، اس		Throughout hist	The foreign conc		
6	5	عبر	להעביר	True Cognate	המנהל החליט י عبر المغامرون النهر		The adventurers	The manager de		
7	6	وحد	אחד	True Cognate	הפרויקט הקהיי وَحَّدَ القائِدُ الشُّجاعُ		The brave leade	The community		
8	7	أعقب	בעקבות	False Friend	בעקבות הגשמי رأى الباحثون مجمو		The researchers	Following the he		

Checklist:

1. Read the Arabic sentence (and its translation if not sure about meaning)
2. Evaluate naturalness: Does it sound "Natural" or "Unnatural" based on the definition above?
3. Add comment if needed
4. Read the Hebrew sentence and its translation
5. Evaluate naturalness: "Natural" or "Unnatural" based on the definition above of naturalness
6. Add optional note if needed
7. *You don't need to verify the correctness of English translation; it is used for easier understanding. There may be issues with translations.*
8. *The English meaning of Arabic and Hebrew words is attached also for reference.*
9. *If you marked the words as cognates and you find that the sentences are awkward (unnatural), please suggest a new sentence without changing the word form.*

2.3 Stage 3: Loanword word-level annotation

Loanwords: the word that has been borrowed from one language (the donor language) and incorporated into the vocabulary of another language (the recipient language). In this case, both Arabic and Hebrew have borrowed the same word (or very similar forms) from a third language.

Label	Definition
Accept	The word pair is correctly identified as a loanword pair with accurate shared meaning. Both forms are used in modern MSA and Modern Hebrew.
Reject	The word pair is NOT a valid loanword pair. Reasons include: not a loanword, archaic usage, proper noun, incorrect meaning, or dialectal usage only.

Google Sheet Columns to Fill:

Column	Required	Description
annotator_type (col. L)	Yes	Accept/Reject
note	Optional	Brief explanation or comment (if needed)

A	B	C	D	E	F	G	H	I	J	K	L
id	source	arabic_wor	hebrew_w	arabic_ipa	hebrew_ipa	arabic_meaning	hebrew_meanin	part_of_s	loan_sourc	phonetic_	annotator_type
1	wiki	جل	ג׳ל	/dʒel/	/ˈdʒel/	gel	gel	noun	English	1	
2	wiki	واط	וואט	/wat/	/wat/	watt	watt	noun	English	1	
3	wiki	زومبي	זומבי	/ˈzombi/	/ˈzombi/	zombie	zombie	noun	English	1	
4	wiki	تايتس	טייץ	/tajts/	/tajts/	tights	tights	noun	English	1	
5	wiki	سوشي	סושי	/ˈsuʃi/	/ˈsuʃi/	sushi	sushi	noun	Japanese	1	
6	wiki	هلو	הלו	/haˈlo/	/haˈlo/	hello	hello	noun	English / In	1	
7	wiki	توش	טוש	/tuʃ/	/tuʃ/	marker, felt-tip p	marker, felt-tip p	noun	English / In	1	
8	wiki	صدمة	טראומה	/ˈtraʊma/	/ˈtraʊma/	shock, trauma	trauma	noun	Greek / Inte	1	
9	wiki	لجيتيمي	לגיטימי	/legiˈtimi/	/legiˈtimi/	legitimate	legitimate	adjective	Latin / Inte	1	
10	wiki	شنيتزل	שניצל	/ʃniˈtsel/	/ʃniˈtsel/	schnitzel	schnitzel	noun	German	1	

How do decide on the loanwords:

1. Verify the word is a genuine loanword, The word should be commonly used in Modern Standard Arabic (MSA) and Modern Hebrew. The word should originate from a non-Semitic source language, Both Arabic and Hebrew forms should be phonologically similar.
2. The provided meaning should be correct for BOTH languages.
3. If meanings differ significantly between Arabic and Hebrew, note this in the comments.

4. Words to REJECT: i) Words that are NOT loanwords (native Semitic words) ii) Archaic words that are rarely used in modern contexts Proper nouns (names of people, places, brands) iii) Words that are only used in dialectal Arabic, not MSA. iv） Words that are completely different in meaning between the two languages
5. You will also see other columns such as entry, id, arabic_ipa, hebrew_ipa, phonetic_similairty, loan_source, those columns are only used for easier understanding by researchers. ***please*** *use your own judgement when providing your annotation!*
6. If you have any comments, add a note in column M.

2.4 Stage 4: Loanword Sentence-level annotation

The requirement is same as stage 2.

3. Strict Guidelines & Critical Rules

Here are some important rules to follow:

1. NO use of LLMs or AI tools (ChatGPT, Claude, Gemini, etc.)
2. NO use of machine translation
3. NO guessing — consult a dictionary if uncertain
4. Verify meanings independently using trusted dictionaries
5. Maintain consistency throughout the dataset
6. Add notes to explain unclear or borderline cases

4. Additional examples

Example 1: True Cognate (Stage 1)

[Input Variables]

arabic_undiac	hebrew_undiac	arabic_meaning	hebrew_meaning
عبد	עבד	slave; servant; worship	slave; to work

[Expected Annotation]

annotator_type	note
True Cognate	Both share core meaning "slave/servant" from root ʕ-b-d

Example 2: False Friend (Stage 1)

[Input Variables]

arabic_undiac	hebrew_undiac	arabic_meaning	hebrew_meaning
عزر	עזרה	reprimand; censure	help; assistance

[Expected Annotation]

annotator_type	note
False Friend	Arabic "reprimand" vs Hebrew "help" — no semantic overlap

Example 3: Natural Hebrew, Unnatural Arabic (Stage 2)

[Input Variables]

arabic_sentence	hebrew_sentence
كان المشتبه به هو الفاعل الرئيسي في عملية السرقة.	החשוד היה הפועל הראשי במעשה השוד.

[Expected Output]

Arabic sentence natural	Hebrew sentence natural	arabic_note	hebrew_note
Unnatural	Natural	The word "الفاعل" is not commonly used to mean "perpetrator" in this context. A more natural choice would be "المُنفِّذ" or rephrase the sentence entirely.	

Example 4: Both Sentences Natural (Stage 2)

Sentences

arabic_sentence	hebrew_sentence
اشترى الراعي عنزة بيضاء لتنتج حليبًا طازجًا كل صباح.	הילד ליטף עז קטנה בחווה והאכיל אותה עשב טרי.

[Expected Annotation]

Arabic sentence natural	Hebrew sentence natural	arabic_note	hebrew_note
Natural	Natural		

Example 5: Unnatural Hebrew (Stage 2)

Sentences

arabic_sentence	hebrew_sentence
عبر الطريق بسرعة.	הוא להעביר את הקבצים מחר.

Expected Annotation

arabic_sentence_natural	hebrew_sentence_natural	arabic_note	hebrew_note
Natural	Unnatural		Missing conjugation; should be "העביר" or "צריך להעביר"

Example 6: Accepted Loanwords (Stage 3)

[Input Variables]

arabic_word	Hebrew_word	arabic_meaning	hebrew_meaning
سوشي	סושי	sushi	sushi

[Expected Annotation]

annotator_type	note
Accept	Common loanword from Japanese, widely used in both languages

Example 7: Rejected Loanwords (Stage 3)

[Input Variables]

arabic_word	Hebrew_word	arabic_meaning	hebrew_meaning

[Expected Annotation]

annotator_type	note

5. Common Mistakes to Avoid

1. Over-reliance on form — Similar-looking words may be False Friends if meanings differ
2. Ignoring polysemy — One shared meaning is sufficient for True Cognate
3. Translation bias — Verify meanings in original languages
4. Conjugation differences are expected and should not be a factor in the annotation decision.

Final Submission Checklist

- All entries have annotator_type filled (Stage 1)
- All entries have arabic_sentence_natural filled (Stage 2)
- All entries have hebrew_sentence_natural filled (Stage 2)
- Notes added for unclear or borderline cases
- No entries skipped

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